ME464 HW 5 (FK Stanford)-1

Sunday, April 5, 2026 9:56 PM

ME464 HW 5 (FK Stan...

Paul Martin
Student Name

HW #5: WRITTEN

Box answers and show all work in the space provided. Submit at the start of class on the scheduled due date.

For the following questions, consider the Stanford Manipulator shown in Figure 1.

The local frames of each link are show as $x_0 y_0 z_0$ for the base link to $x_6 y_6 z_6$ for end-effector link 6. The joint axis are about the z axis of the previous link, such that $\boldsymbol{\omega}_1 = z_0$, $\boldsymbol{\omega}_2 = z_1$, etc. Each axis is a revolute joint, except for $\boldsymbol{\omega}_3$, which is prismatic. The distances d_{1-6} give the dimensions of the robot.

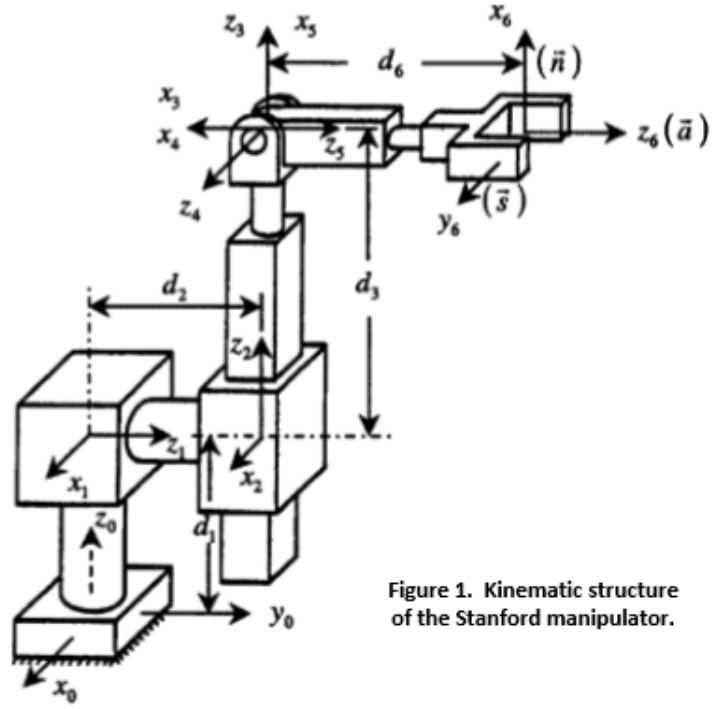

Figure 1. Kinematic structure of the Stanford manipulator.

1) Find the home position, $\mathbf{g}_{st}(0)$, of the tool frame (link 6).

$$g_{st}(0) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 + d_6 \\ 1 & 0 & 0 & d_1 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2) Find each joint axis, $\boldsymbol{\omega}_{1-6}$.

$$\omega_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \omega_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \omega_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\omega_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \omega_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \omega_6 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3) Find a point on each joint axis, $\mathbf{q}_{1-6}$.

$$q_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad q_2 = \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix}$$

$$q_4 = \begin{bmatrix} 0 \\ 0 \\ d_2 \end{bmatrix}, \quad q_5 = \begin{bmatrix} 0 \\ d_2 \\ d_1 + d_3 \end{bmatrix}, \quad q_6 = \begin{bmatrix} 0 \\ 0 \\ d_1 + d_3 \end{bmatrix}$$

4) Find the twists, ξ_{1-6}, describing the rigid motion of the each link of the Stanford manipulator.

$$V_1 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ d_1 \end{bmatrix} = \begin{bmatrix} -d_1 \\ 0 \\ 0 \end{bmatrix}$$

$$V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V_4 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ d_2 \\ 0 \end{bmatrix} = \begin{bmatrix} d_2 \\ 0 \\ 0 \end{bmatrix}$$

$$V_5 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ d_2 \\ 0 \end{bmatrix} = \begin{bmatrix} d_2 \\ 0 \\ 0 \end{bmatrix}$$

$$V_6 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ d_1 + d_3 \end{bmatrix} = \begin{bmatrix} -d_1 - d_3 \\ 0 \\ 0 \end{bmatrix}$$

$$\xi_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \xi_2 = \begin{bmatrix} -d_1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \xi_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xi_4 = \begin{bmatrix} d_2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \xi_5 = \begin{bmatrix} 0 \\ d_1 + d_3 \\ -d_2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \xi_6 = \begin{bmatrix} -d_1 - d_3 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$